Lab Experiment: Comparison of Logistic Regression and KNN Classifiers

Aim: To implement Logistic Regression and K Nearest Neighbors (KNN) classifiers on the Breast Cancer Wisconsin (Diagnostic) dataset and compare their performance using standard classification evaluation metrics.

1. Import libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f
```

2. Load the dataset

The dataset file wdbc.data has no header row. Column names are added based on the wdbc.names file.

In [2]:
```
features = ['radius', 'texture', 'perimeter', 'area', 'smoothness', 'compact
            'concavity', 'concave_points', 'symmetry', 'fractal_dimension']

columns = ['id', 'diagnosis'] + [f'{stat}_{f}' for stat in ['mean', 'se', 'w

df = pd.read_csv('data/wdbc.data', header=None, names=columns)
df.head()
```

Out[2]:

	id	diagnosis	mean_radius	mean_texture	mean_perimeter	mean_area	m
0	842302	M	17.99	10.38	122.80	1001.0	
1	842517	M	20.57	17.77	132.90	1326.0	
2	84300903	M	19.69	21.25	130.00	1203.0	
3	84348301	M	11.42	20.38	77.58	386.1	
4	84358402	M	20.29	14.34	135.10	1297.0	

5 rows × 32 columns

In [3]:
```
df.shape
```

Out[3]:
```
(569, 32)
```

Dataset has 569 rows and 32 columns (id, diagnosis and 30 numeric features).

3. Exploratory Data Analysis

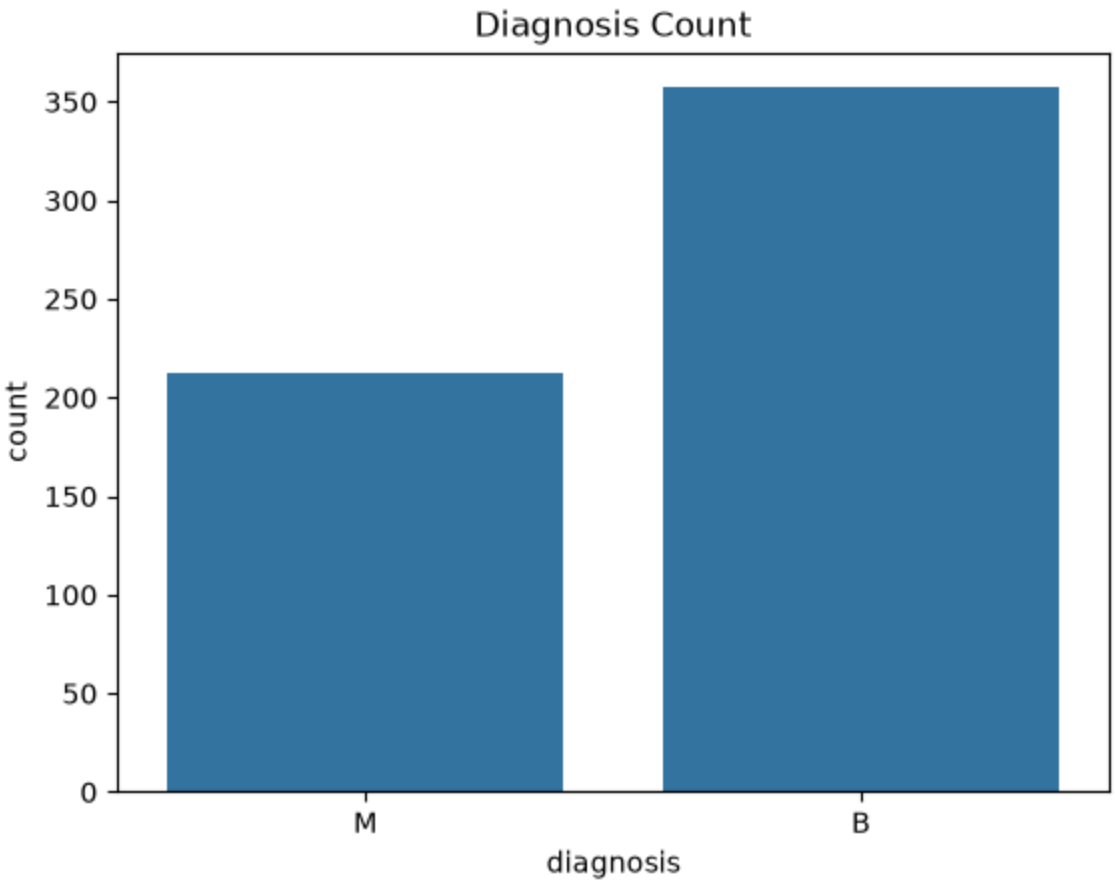
In [4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 32 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   id                       569 non-null    int64
 1   diagnosis                569 non-null    str
 2   mean_radius              569 non-null    float64
 3   mean_texture             569 non-null    float64
 4   mean_perimeter           569 non-null    float64
 5   mean_area                569 non-null    float64
 6   mean_smoothness          569 non-null    float64
 7   mean_compactness         569 non-null    float64
 8   mean_concavity           569 non-null    float64
 9   mean_concave_points      569 non-null    float64
 10  mean_symmetry            569 non-null    float64
 11  mean_fractal_dimension   569 non-null    float64
 12  se_radius                569 non-null    float64
 13  se_texture               569 non-null    float64
 14  se_perimeter             569 non-null    float64
 15  se_area                  569 non-null    float64
 16  se_smoothness            569 non-null    float64
 17  se_compactness           569 non-null    float64
 18  se_concavity             569 non-null    float64
 19  se_concave_points        569 non-null    float64
 20  se_symmetry              569 non-null    float64
 21  se_fractal_dimension     569 non-null    float64
 22  worst_radius             569 non-null    float64
 23  worst_texture            569 non-null    float64
 24  worst_perimeter          569 non-null    float64
 25  worst_area               569 non-null    float64
 26  worst_smoothness         569 non-null    float64
 27  worst_compactness        569 non-null    float64
 28  worst_concavity          569 non-null    float64
 29  worst_concave_points     569 non-null    float64
 30  worst_symmetry           569 non-null    float64
 31  worst_fractal_dimension  569 non-null    float64
dtypes: float64(30), int64(1), str(1)
memory usage: 142.4 KB
```

In [5]:
```
df['diagnosis'].value_counts()
```

Out[5]:
```
diagnosis
B    357
M    212
Name: count, dtype: int64
```

In [6]:
```
sns.countplot(x='diagnosis', data=df)
plt.title('Diagnosis Count')
plt.show()
```



Out of 569 patients, 357 are benign (B) and 212 are malignant (M). The classes are not equal but not too imbalanced either.

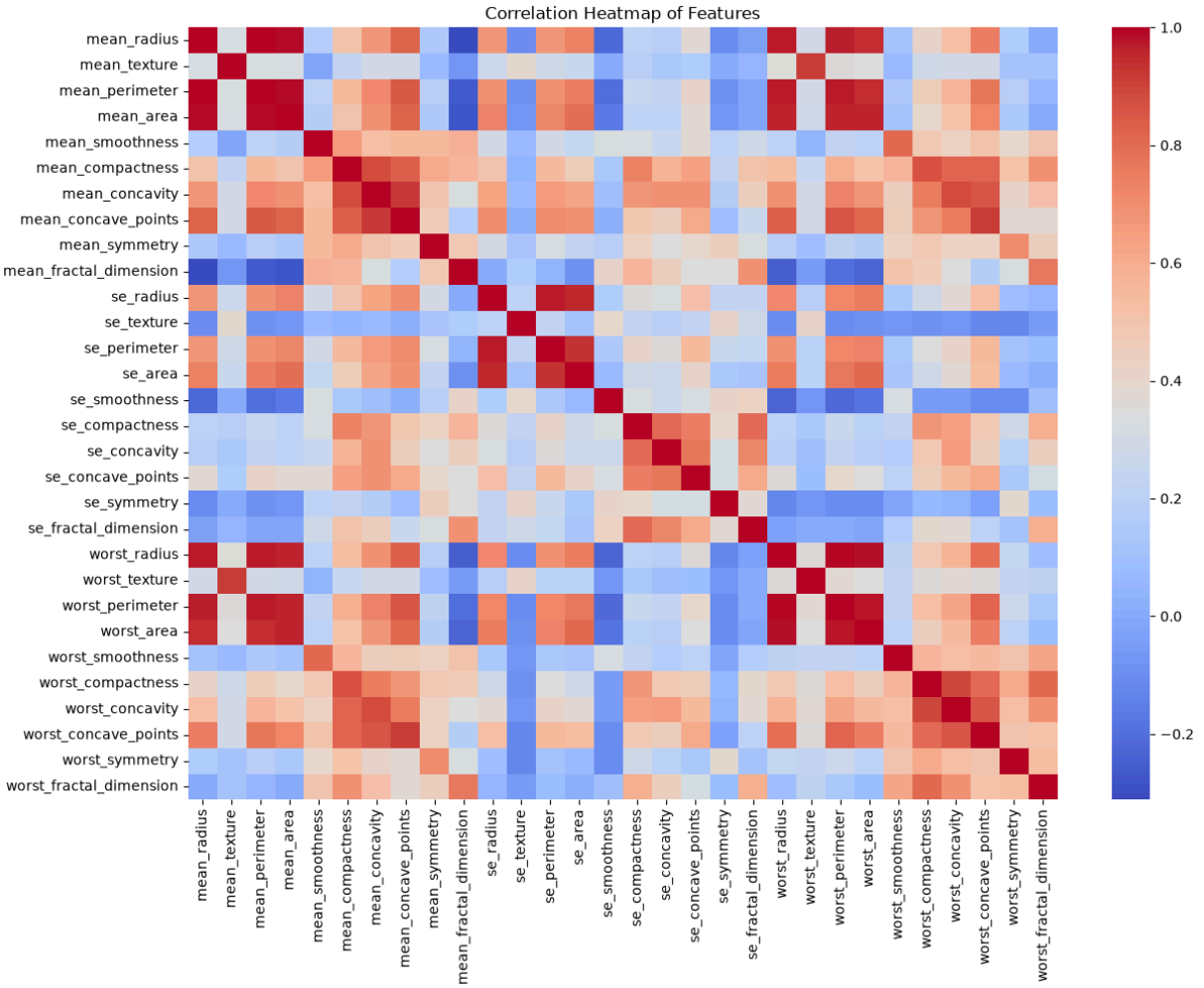
In [7]:
```
df.describe()
```

Out[7]:

	id	mean_radius	mean_texture	mean_perimeter	mean_area	mea
count	5.690000e+02	569.000000	569.000000	569.000000	569.000000	
mean	3.037183e+07	14.127292	19.289649	91.969033	654.889104	
std	1.250206e+08	3.524049	4.301036	24.298981	351.914129	
min	8.670000e+03	6.981000	9.710000	43.790000	143.500000	
25%	8.692180e+05	11.700000	16.170000	75.170000	420.300000	
50%	9.060240e+05	13.370000	18.840000	86.240000	551.100000	
75%	8.813129e+06	15.780000	21.800000	104.100000	782.700000	
max	9.113205e+08	28.110000	39.280000	188.500000	2501.000000	

8 rows × 31 columns

In [8]:
```
plt.figure(figsize=(14, 10))
sns.heatmap(df.drop(['id', 'diagnosis'], axis=1).corr(), cmap='coolwarm')
plt.title('Correlation Heatmap of Features')
plt.show()
```



Many of the mean, se and worst versions of the same feature are highly correlated with each other, like radius, perimeter and area. This is expected since they are derived from the same cell shape.

4. Check missing values

In [9]:
```
df.isnull().sum().sum()
```

Out[9]:
```
np.int64(0)
```

There are no missing values in the dataset, so no imputation is needed.

5. Data preprocessing

In [10]:
```
df = df.drop('id', axis=1)
df['diagnosis'] = df['diagnosis'].map({'M': 1, 'B': 0})
df['diagnosis'].value_counts()
```

Out[10]:
```
diagnosis
0    357
1    212
Name: count, dtype: int64
```

id column is dropped since it has no predictive value. diagnosis is encoded as 1 for malignant and 0 for benign.

In [11]:
```
X = df.drop('diagnosis', axis=1)
y = df['diagnosis']
```

6. Train test split

In [12]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, rar
X_train.shape, X_test.shape
```

Out[12]:
```
((455, 30), (114, 30))
```

Data is split into 80% training and 20% testing. stratify=y keeps the same ratio of benign to malignant in both sets.

7. Feature scaling

KNN uses distance between points so features need to be on the same scale. Scaling is applied for both models to keep comparison fair.

In [13]:
```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
```

```
X_test_scaled = scaler.transform(X_test)
```

8. Train Logistic Regression classifier

In [14]:
```
log_reg = LogisticRegression(max_iter=10000)
log_reg.fit(X_train_scaled, y_train)
y_pred_lr = log_reg.predict(X_test_scaled)
```

9. Train KNN classifier

In [15]:
```
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)
y_pred_knn = knn.predict(X_test_scaled)
```

10. Evaluation

In [16]:
```
def get_metrics(y_true, y_pred):
    return {
        'Accuracy': accuracy_score(y_true, y_pred),
        'Precision': precision_score(y_true, y_pred),
        'Recall': recall_score(y_true, y_pred),
        'F1 Score': f1_score(y_true, y_pred)
    }

lr_metrics = get_metrics(y_test, y_pred_lr)
knn_metrics = get_metrics(y_test, y_pred_knn)

lr_metrics
```

Out[16]:
```
{'Accuracy': 0.9649122807017544,
 'Precision': 0.975,
 'Recall': 0.9285714285714286,
 'F1 Score': 0.9512195121951219}
```

In [17]:
```
knn_metrics
```

Out[17]:
```
{'Accuracy': 0.956140350877193,
 'Precision': 0.9743589743589743,
 'Recall': 0.9047619047619048,
 'F1 Score': 0.9382716049382716}
```

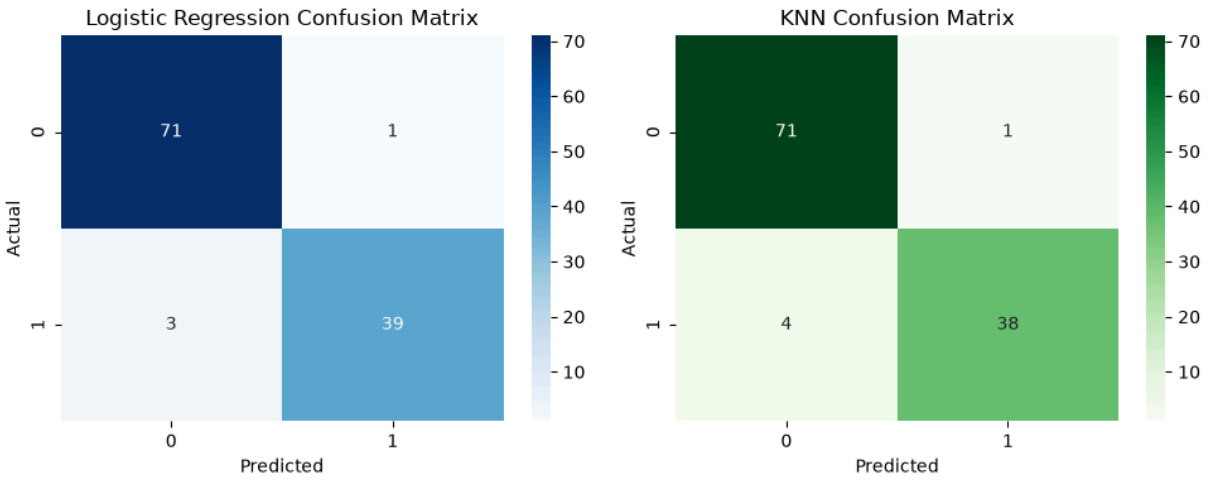
In [18]:
```
cm_lr = confusion_matrix(y_test, y_pred_lr)
cm_knn = confusion_matrix(y_test, y_pred_knn)

fig, axes = plt.subplots(1, 2, figsize=(10, 4))
sns.heatmap(cm_lr, annot=True, fmt='d', cmap='Blues', ax=axes[0])
axes[0].set_title('Logistic Regression Confusion Matrix')
axes[0].set_xlabel('Predicted')
axes[0].set_ylabel('Actual')

sns.heatmap(cm_knn, annot=True, fmt='d', cmap='Greens', ax=axes[1])
```

```
axes[1].set_title('KNN Confusion Matrix')
axes[1].set_xlabel('Predicted')
axes[1].set_ylabel('Actual')

plt.tight_layout()
plt.show()
```



Logistic Regression made 4 wrong predictions out of 114 test samples (1 false positive, 3 false negatives). KNN made 5 wrong predictions (1 false positive, 4 false negatives). Both models are confusing malignant cases as benign more than the other way around.

11. Comparison table

In [19]:
```
comparison = pd.DataFrame([lr_metrics, knn_metrics], index=['Logistic Regres
comparison
```

Out[19]:

	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.964912	0.975000	0.928571	0.951220
KNN	0.956140	0.974359	0.904762	0.938272

12. Interpretation

Logistic Regression got 96.5% accuracy and KNN got 95.6% accuracy on the test set. Logistic Regression is also slightly better in recall (0.93 vs 0.90) and F1 score (0.95 vs 0.94), while precision is almost the same for both (about 0.97).

Recall matters a lot in this problem because missing a malignant case (false negative) is more dangerous than a false alarm. Logistic Regression had 3 false negatives compared to 4 for KNN, so it is doing slightly better on this front too.

Overall Logistic Regression performs a bit better than KNN on this dataset across all four metrics. This makes sense because the classes are described in the dataset

documentation as linearly separable, which favors a linear model like Logistic Regression.